

HONEYWELL SOFTWARE ENGINEERING TEAMS

AI Champions Programme

Module 08

PR Process Automation, Quality Gates and LLM-as-Judge

Configuring AI-assisted PR review against specification compliance, coding standards, and test evidence — then adding LLM-as-Judge scoring, evidence-based comments, and a clear human escalation path.

	DURATION		FORMAT		DELIVERED FOR
	1.5 Hours		Hands-On Lab		Honeywell Eng. Teams

How We'll Spend These Ninety Minutes

Six connected moves — from what a PR review checks to a quality gate that's actually been tested.

01

From Test Evidence to PR Quality Gates

10 MIN

Why PR review is where Module 06 and 07's test evidence and governed tools actually get used.

02

What AI-Assisted PR Review Checks

20 MIN

Specification compliance, coding/design standards, security, test evidence, and regression risk.

03

LLM-as-Judge: Evaluation Rubrics & Confidence

20 MIN

How a PR diff becomes a scored verdict, and what happens when confidence is low.

04

Evidence-Based Comments & Human Escalation

15 MIN

What separates a useful review comment from a vague one, and when a human needs to step in.

05

Review-Cycle Reduction: Where Time Actually Goes

10 MIN

Comparing a manual review pass to an AI-assisted one, stage by stage.

06

Hands-On Lab: Configure & Run a PR Quality Flow

15 MIN

Run the flow against a deliberately flawed change and close the loop with human approval.

Module 08 in the 12-Module Journey

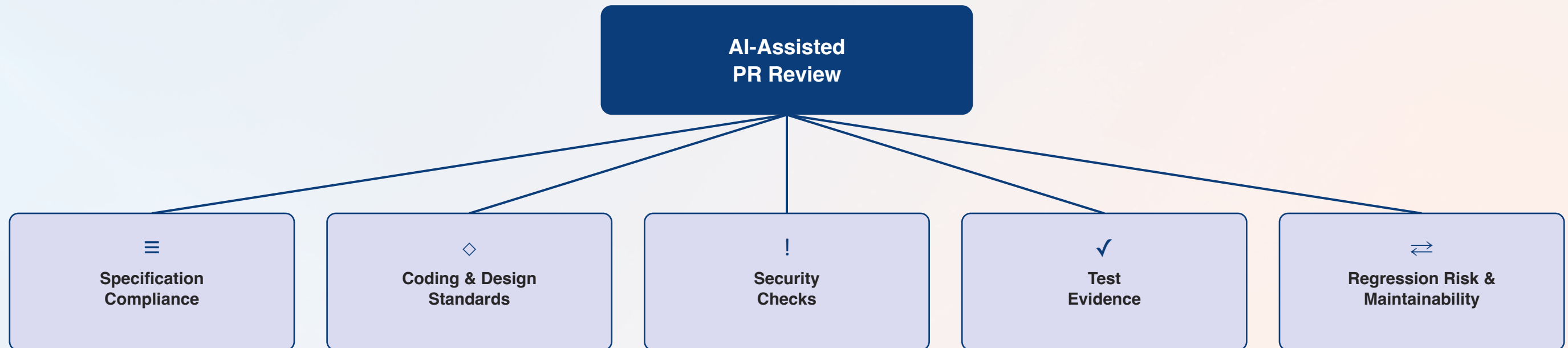
The test evidence Module 06 built and the governed tools Module 07 connected now meet at the PR.



Why this matters: Module 09's studio workflows and Module 10's Agent Prism both assume PRs already carry this kind of evidence trail — this module is where that trail starts.

AI-Assisted PR Review: What Gets Checked

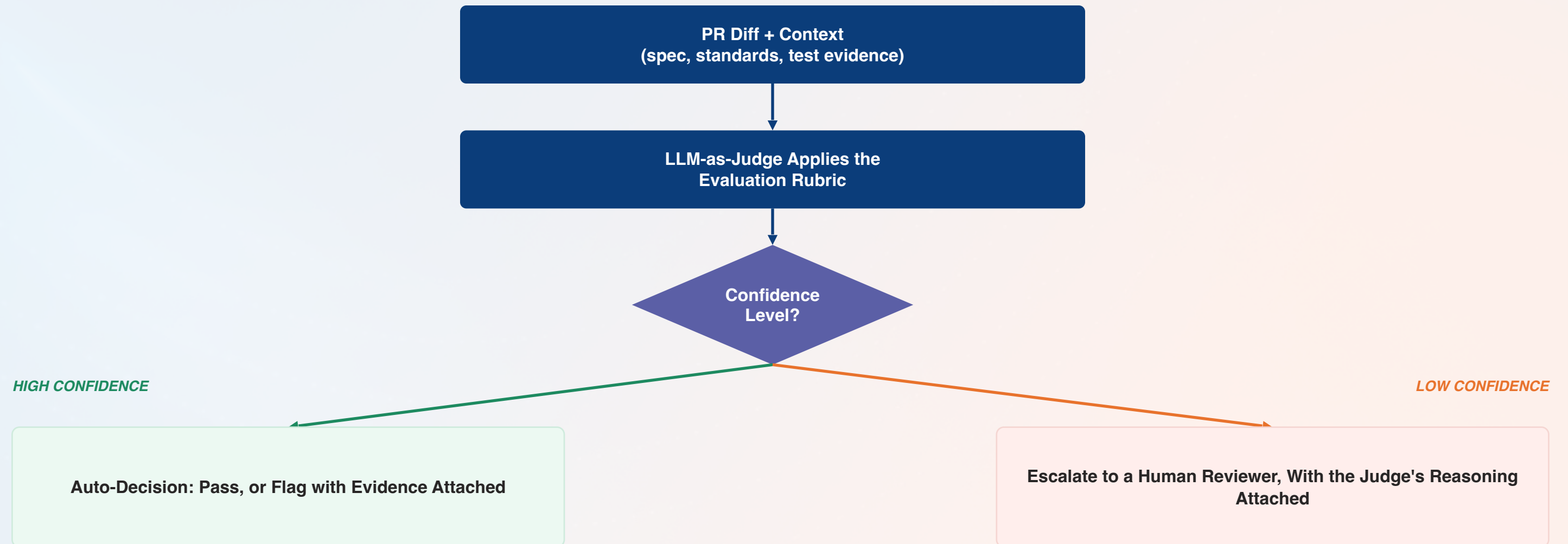
Five categories, checked consistently on every PR — not whichever the reviewer remembers this time.



Why this matters: Every one of these traces back to earlier modules — the spec from Module 04, the standards from Module 05, the tests from Module 06.

LLM-as-Judge: From PR Diff to Verdict

A scored evaluation against a rubric, branching on how confident the judge actually is.



Evidence-Based Comments, Not Vague Ones

The difference isn't tone — it's whether the comment points to something specific and checkable.

EVIDENCE-BASED

“Line 42 violates AC-03 (timeout must return 503) — spec §4.2”

“test_duplicate_order fails against Testcontainers Postgres — CI run #128”

“Duplicates retry logic in PaymentClient.java:88 — consider extracting a helper”

“No test covers the negative case in AC-05 — coverage gap”

VAGUE

“This looks wrong.”

“Please fix the tests.”

“Consider refactoring this.”

“Not sure this is safe.”

Review-Cycle Reduction: Where Time Actually Goes

Not “AI reviews faster” in the abstract — specific stages, specifically changed.

Stage	Manual Review	AI-Assisted Review
Initial read-through	Human reads the full diff	AI summarizes the diff and flags key changes
Standards check	Manual checklist walk-through	Automated rubric check against coding/design standards
Test evidence check	Human verifies tests ran	Evidence auto-attached from CI and Testcontainers runs
Escalation	Every PR needs full human review	Only low-confidence or flagged PRs need full human review

The savings come from narrowing full human review to the PRs that actually need it — not from skipping review.

Hands-On Lab Preview: Configure & Run a PR Quality Flow

The 15-minute activity that closes this module.



**Configure the
Quality Flow**

Wire up the PR template, quality checklist, and LLM-as-Judge rubric.



**Run AI
Review**

Against a deliberately flawed change, to see what it actually catches.



**Run
LLM-as-Judge**

Score the change against the rubric and assign a confidence level.



**Capture
Evidence**

Record what was checked, what passed, and what didn't.



**Close the
Loop**

Human approval on flagged items, comparing time and quality to a manual pass.

The throughline: A quality gate that never flags anything hasn't been tested — this lab makes sure yours actually catches the flaw.

Module 08 Outcomes & What's Next

- ✓ What AI-assisted PR review checks — spec, standards, security, tests, regression risk — is understood
- ✓ LLM-as-Judge's evaluation flow, from PR diff to verdict, is clear, confidence branching included
- ✓ Evidence-based review comments, and what separates them from vague ones, are understood
- ✓ Where review-cycle time is actually saved, stage by stage, is clear
- ✓ A PR quality flow has been configured and run against a deliberately flawed change
- ✓ The loop has been closed with human approval on flagged items



NEXT MODULE

Module 09 — Studio-Style Workflows for Business and Engineering Collaboration (2 hours). How non-technical program and product roles author agent-assisted workflows in Atlassian Rovo Studio, grounded on the same governed MCP connections built in Module 07.

Questions & Discussion

Module 08 — PR Process Automation, Quality Gates and LLM-as-Judge